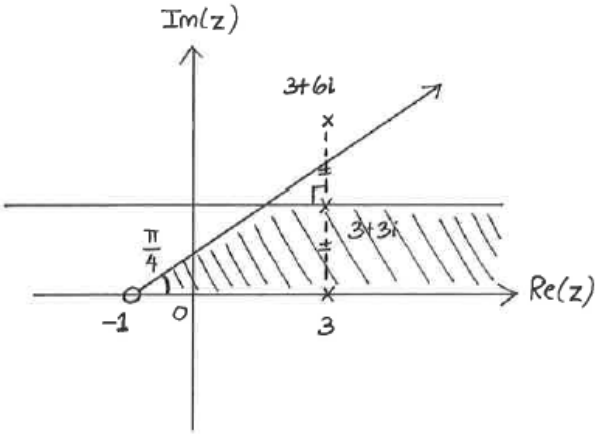
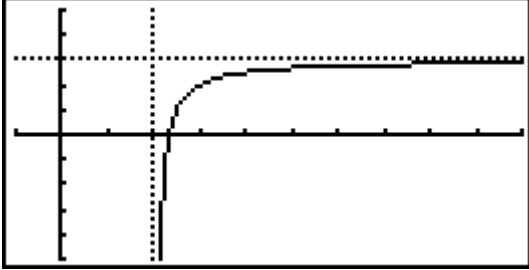


2014 MI H2 Mathematics Paper 2 Answer

1 (i)	$\frac{-x^2 + 4x + 2}{x + 4} \geq 1$ $\frac{x^2 - 3x + 2}{x + 4} \leq 0$ $\frac{(x-1)(x-2)}{x+4} \leq 0$ $x < -4 \text{ or } 1 \leq x \leq 2$
1 (ii)	$\frac{-2\sin^2 \theta + 4\sin \theta + 1}{\sin \theta + 2} \geq 1$ $\frac{-4\sin^2 \theta + 8\sin \theta + 2}{2\sin \theta + 4} \geq 1$ $\frac{-(2\sin \theta)^2 + 4(2\sin \theta) + 2}{(2\sin \theta) + 4} \geq 1$ $2\sin \theta < -4 \text{ or } 1 \leq 2\sin \theta \leq 2$ No solutions $\frac{\pi}{6} \leq \theta \leq \frac{\pi}{2}$
2	$10000 = x^2 h$ $h = \frac{10000}{x^2}$ Let C be the total cost in dollars. $C = 5x^2 + 2(4xh)$ $C = 5x^2 + \frac{80000}{x}$ $\frac{dC}{dx} = 10x - \frac{80000}{x^2}$ Let $\frac{dC}{dx} = 0$, $x = 20$ $\frac{d^2C}{dx^2} = 10 + \frac{160000}{x^3}$ $\left. \frac{d^2C}{dx^2} \right _{x=20} = 30 > 0$ Hence, C is minimum when $x = 20$. $C_{\min} = 6000$

3 (i)	
3 (ii)	$\sqrt{[2 - (-2)]^2 + (3 - 7)^2}$ $= \sqrt{32}$
3 (iii)	$-\tan^{-1} 3 < \arg(z - (-2 + 3i)) \leq 0$
4 (i)	$\frac{dy}{dx} = \frac{1}{2t}$ Equation of tangent at the point with parameter p: $y - \left(2 - \frac{1}{2p}\right) = \frac{1}{2p}(x - \ln p)$ $y = \frac{x}{2p} - \frac{\ln p}{2p} - \frac{1}{2p} + 2$
4 (ii)	Equation of tangent at the A (i.e. $p = 1$): $y = \frac{x}{2} + \frac{3}{2}$ When $y = 0$, $x = -3$ $T(-3, 0)$ Equation of normal at A: $y = -2x + \frac{3}{2}$ When $y = 0$, $x = \frac{3}{4}$ $N\left(\frac{3}{4}, 0\right)$ Area of $\Delta ANT = \frac{45}{16}$

5 (a) (i)	 Asymptotes: $x = 2$, $y = 3$ Axial intercepts: $\left(\frac{7}{3}, 0\right)$
5 (a) (ii)	$f^{-1}(x) = 2 - \frac{1}{x-3}$, $x < 3$, $x \in \mathbb{R}$
5 (a) (iii)	$R_g = \left(\frac{1}{2-a} + 3, 3\right)$ Let $\frac{1}{2-a} + 3 = 2$ $a = 3$
5 (a) (iv)	$fg(x) = \frac{2-x}{x-3} + 3$, $x > 3$, $x \in \mathbb{R}$ $R_{fg} = (-\infty, 2)$
5 (b)	$f(x) = \frac{1}{2} \left(1 - \frac{x}{2}\right)^{-1} + 3$ $= \frac{1}{2} \left[1 + (-1) \left(-\frac{x}{2}\right) + \frac{(-1)(-2)}{2!} \left(-\frac{x}{2}\right)^2 + \frac{(-1)(-2)(-3)}{3!} \left(-\frac{x}{2}\right)^3 + \dots \right] + 3$ $= \frac{7}{2} + \frac{x}{4} + \frac{x^2}{8} + \frac{x^3}{16} + \dots$
6 (i)	A random sample is a sample where every employee has an equal chance of being selected and each employee is selected independently of each other.
6 (ii)	Assign each employee a number from 1 to 1000. Use a random number generator to generate 100 distinct numbers between 1 and 1000 (inclusive). Select the employees with the randomly generated numbers.

6 (iii)	Stratified sampling. The number of employees in the sample is proportional to the number of employees that work in the different shifts and hence the sample will give a better representation of all the employees.
7 (i)	Let S_n be the random variable for the number of students who stop to watch the concert in an n -minute interval. $S_3 \sim \text{Po}(18)$ $P(S_3 \geq 15) = 1 - P(S_3 \leq 14) = 0.79192$ $= 0.792$ (3 s.f.)
7 (ii)	Let L_t be the random variable for the number of students who leave the concert venue in a t -minute interval. $L_t \sim \text{Po}(3t)$ $P(L_t \geq 1) = 0.6$ $P(L_t = 0) = 0.4$ $e^{-3t} = 0.4$ $t = 0.30543$ (5 d.p.)
7 (iii)	$S_5 \sim \text{Po}(30)$ $L_5 \sim \text{Po}(15)$ Since $30 > 10$, $S_5 \sim N(30, 30)$ approximately Since $15 > 10$, $L_5 \sim N(15, 15)$ approximately $S_5 - L_5 \sim N(15, 45)$ $P(10 \leq S_5 - L_t < 20)$ c.c. $= P(9.5 \leq S_5 - L_5 < 19.5)$ $= 0.543$ (3 s.f.)
8 (i)	$\bar{x} = 10.2$ $s^2 = 0.031667 = 0.0317$ (3 s.f.)
8 (ii)	Assume X follows a normal distribution. $H_0 : \mu = 10$ $H_1 : \mu \neq 10$ Perform a 2-tailed t -test. Test statistic, $t = \frac{\bar{X} - \mu}{s / \sqrt{n}} \sim t(12)$ where $s = \sqrt{0.031667}$, $n = 13$ $t = 4.0523$ $p\text{-value} = 0.0016038 < 0.05 = \alpha$

	Reject H_0 . There is sufficient evidence at the 5% significance level to conclude that the mean amount of rice in each packet is no longer 10 kg.
8 (iii)	$\alpha = 0.160$ (3 s.f.)
9 (i)	$[P(-1 < X < 1)][P(-1 < Y < 1)]$ $= 0.099988$ (5 s.f.)
9 (ii)	$[P(-5 < X < 5)][P(-5 < Y < 5)] - 0.099988$ $= 0.893187 - 0.099988$ $= 0.793199 = 0.793$ (3 s.f.)
9 (iii)	$\frac{0.793199}{0.893187}$ $= 0.888$ (3 s.f.)
9 (iv)	$(0.793199)^2 + 2(0.099988)(1 - 0.893187)$ $= 0.651$ (3 s.f.)
10 (i)	
10 (ii)	Model (A): $r = 0.867$ Model (B): $r = -0.999$ For model (B), $ r $ is closer to 1 compared to model (A). Hence, model (B) is more appropriate. OR As t increases, the values of h increase at a decreasing rate. Hence, model (B) is more appropriate.
10 (iii)	Regression line: $h = -20.437\frac{1}{t} + 25.642$ When $h = 22$, $t = 5.61$ (3 s.f.)
10 (iv)	a remains unchanged. b will increase by k .
11 (i)	$\left(\frac{26}{36}\right)^4 = 0.272$ (3 s.f.)

11 (ii)	$\frac{9+8+7+6+5+4+3+2+1}{36^2} = 0.0347 \text{ (3 s.f.)}$
11 (iii)	$\frac{(26)(25)(24)(10)({}^4C_1)}{36^4} = 0.372 \text{ (3 s.f.)}$
11 (iv)	$\frac{(26)(10)(9)({}^4C_2) + (10)(26)(25)({}^4C_2)}{36^4}$ $= 0.0316 \text{ (3 s.f.)}$
12 (i)	Let S be the random variable for the lifetime of a Standard battery in hours. $S \sim N(300, 30^2)$ $P(S > 310) = 0.369 \text{ (3 s.f.)}$
12 (ii)	Let L be the random variable for the lifetime of a Long Lasting battery in hours. $L \sim N(500, 50^2)$ $S_1 + \dots + S_{10} - 5L \sim N(500, 71500)$ $P(S_1 + \dots + S_{10} - 5L < 0) = 0.0307 \text{ (3 s.f.)}$
12 (iii)	Let T be the random variable for the number of torchlights out of 10 that are in working condition. $T \sim B(10, 0.97)$ $P(T \geq 9) = 1 - P(T \leq 8)$ $= 0.965493 = 0.965 \text{ (3 s.f.)}$
12 (iv)	Let U be the random variable for the number of boxes out of 15 that have at least 9 torchlights that are in working condition. $U \sim B(15, 0.965493)$ $P(12 < U < 15) = P(U \leq 14) - P(U \leq 12)$ $= 0.396 \text{ (3 s.f.)}$
12 (v)	Let V be the random variable for the number of torchlights out of 100 that are spoilt. $V \sim B(100, 0.03)$ $n = 100 \geq 50$ is large $np = 3 < 5$ $V \sim \text{Po}(3)$ approximately $P(V \geq 5) = 1 - P(V \leq 4) = 0.185 \text{ (3 s.f.)}$
12 (vi)	$S + L \sim N(800, 3400)$ $(0.97)[P(S + L > 750)] = 0.780 \text{ (3 s.f.)}$

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